# Chapter 4 solutions

## DM04-E01

At $n = 1$, both sides equal one. Assume the formula for $k$. Then

$$\sum_{i=1}^{k+1} i^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{(k+1)(k(2k+1)+6(k+1))}{6}.$$

The remaining factor is $2k^2 + 7k + 6 = (k+2)(2k+3)$, giving

$$\frac{(k+1)(k+2)(2k+3)}{6},$$

the formula at $k+1$. Induction completes the proof.

## DM04-E02

The strong-induction base two is a prime, hence a one-prime product. For arbitrary $n > 2$, if prime it already factors. Otherwise choose divisor $1 < a < n$ and put $b = n/a$, with $1 < b < n$. The hypothesis factors both smaller integers into primes; combining their factors gives a prime factorization of $n$.

## DM04-E03

The required quotient is $q = \lfloor -37/8 \rfloor = -5$, with

$$-37 = 8(-5) + 3,$$

so $r = 3$. C99 and C++11 onward specify integer division with the quotient truncated toward zero, giving quotient minus four and `-37 % 8` equal to minus five, since $-37 = 8(-4) - 5$. If a language returns a negative remainder while the problem requires a nonnegative one, add the positive divisor to $r$ and subtract one from $q$.

## DM04-E04

$$1071 = 2 \cdot 462 + 147, \quad 462 = 3 \cdot 147 + 21, \quad 147 = 7 \cdot 21.$$

The gcd is 21. Back-substitution gives

$$21 = 462 - 3 \cdot 147 = 462 - 3(1071 - 2 \cdot 462) = 7 \cdot 462 - 3 \cdot 1071.$$

Thus coefficients are $s = -3, t = 7$.

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## DM04-E05

No. Induction needs both a starting assertion and a successor implication. The step only says that if one term is true, the next is true. Without a starting point, no term follows. The implications for this particular true formula are not vacuous because their antecedents are true. For a counterexample, let $Q(n) : n < 0$ for integers $n \geq 1$. Every $Q(k)$ is false, so every $Q(k) \to Q(k+1)$ is true, while the universal conclusion is false. The step alone cannot establish a starting point.

## DM04-E06

Induct on $m$. For $m = 1$, the premise itself is the conclusion. Assume the claim for $m = k$. If $p \mid (a_1 \cdots a_k)a_{k+1}$, Euclid's lemma gives $p \mid a_1 \cdots a_k$ or $p \mid a_{k+1}$. In the first case the induction hypothesis supplies an index among the first $k$; in the second choose $k+1$.

## DM04-E07

At one, $1^3 = (1 \cdot 2/2)^2 = 1$. Assume $\sum_{i=1}^{k} i^3 = (k(k+1)/2)^2$, for fixed $k \geq 1$. Add the next term:

$$\begin{aligned}\sum_{i=1}^{k+1} i^3 &= \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 \\ &= (k+1)^2\left(\frac{k^2}{4} + k + 1\right) \\ &= (k+1)^2\frac{k^2+4k+4}{4} \\ &= \left(\frac{(k+1)(k+2)}{2}\right)^2.\end{aligned}$$

This is the next-index formula, proving the claim by induction.

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## DM04-E08

The gcd is 21, and $21 = 1071(-3) + 462(7)$. Since $42 = 2 \cdot 21$, one solution is $(x_0, y_0) = (-6, 14)$. Substitution checks $-6426 + 6468 = 42$.

Subtract its equation from that of another solution:

$$1071(x + 6) + 462(y - 14) = 0,$$
$$51(x + 6) + 22(y - 14) = 0.$$

Coprimality of 51 and 22 forces $x + 6 = 22k, y - 14 = -51k$. Therefore every integer solution is

$$x = -6 + 22k, \qquad y = 14 - 51k, \qquad k \in \mathbb{Z}.$$

Conversely, substitution cancels the added terms, so all these pairs solve the equation.

## DM04-E09

$$b_2 = 5 \cdot 5 - 6 \cdot 2 = 13, \quad b_3 = 5 \cdot 13 - 6 \cdot 5 = 35, \quad b_4 = 5 \cdot 35 - 6 \cdot 13 = 97.$$

The candidate gives $3^0 + 2^0 = 2$, $3^1 + 2^1 = 5$, and $3^2 + 2^2 = 13$. This induction step uses two preceding terms, so this proof verifies the base cases zero and one. For $k \geq 1$, assume the formula at $k, k - 1$. Then

$$\begin{aligned} b_{k+1} &= 5(3^k + 2^k) - 6(3^{k-1} + 2^{k-1}) \\ &= (5 \cdot 3 - 6)3^{k-1} + (5 \cdot 2 - 6)2^{k-1} \\ &= 9 \cdot 3^{k-1} + 4 \cdot 2^{k-1} = 3^{k+1} + 2^{k+1}. \end{aligned}$$

Strong induction proves it for all $n \geq 0$.

## DM04-E10

Independent exponent choices in unique factorization give

$$\tau(N) = (4 + 1)(3 + 1)(2 + 1)(1 + 1) = 120.$$

An odd divisor has exponent zero for two, while its exponents for three, five, seven range from zero through 3,2,1. Thus there are $4 \cdot 3 \cdot 2 = 24$ odd divisors.

A square requires even exponents: two permits 0,2,4; three permits 0,2; five permits 0,2; seven permits only zero. Thus square divisors number $3 \cdot 2 \cdot 2 \cdot 1 = 12$.

## DM04-E11

Multiplication gives

$$(a+b\sqrt{-5})(c+d\sqrt{-5}) = (ac-5bd)+(ad+bc)\sqrt{-5}.$$

Expanding squares proves

$$(ac-5bd)^2+5(ad+bc)^2 = (a^2+5b^2)(c^2+5d^2),$$

so the norm is multiplicative. A unit and its inverse have positive integer norms multiplying to one, so each has norm one. This forces $b=0, a=\pm 1$, and conversely these two elements are units.

If $3=uv$ with neither factor a unit, positive integer norms multiply to nine and must both be three. But $a^2+5b^2=3$ is impossible over integers. Thus one factor is a unit and three is irreducible.

Meanwhile $(2+\sqrt{-5})(2-\sqrt{-5})=9$ is divisible by three, but neither factor has both coefficients divisible by three. Three is therefore not prime. Two has norm four and $\sqrt{-5}$ norm five; both are below nine and neither is a unit. The irreducibility argument must exclude norm three specifically, not every smaller nonunit norm.

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## DM04-E12

The gcds are $\gcd(0,-12)=12$ and $\gcd(-18,0)=18$, since positive common divisors are exactly positive divisors of the nonzero input. If $c=0$, every integer pair solves $0x+0y=c$; otherwise none does. The convention here does not define $\gcd(0,0)$.

For inputs not both zero, let $g=\gcd(a,b)>0$, $A=a/g$, $B=b/g$, with $\gcd(A,B)=1$. Subtracting a particular solution gives

$$A(x-x_0)=-B(y-y_0).$$

If both $A,B\neq 0$, coprime cancellation from Bézout gives $B\mid(x-x_0)$. Write $x-x_0=Bk$, then $y-y_0=-Ak$. Thus

$$x=x_0+(b/g)k,\qquad y=y_0-(a/g)k,\qquad k\in\mathbb{Z}.$$

Substitution cancels the new terms, proving sufficiency too. If $a=0$, then $B=\pm 1$, so the equation fixes $y$ and leaves $x$ free; the same formula gives $x=x_0\pm k, y=y_0$. If $b=0$, it fixes $x$ and lets $y$ vary freely. Thus the formula covers all zero-coordinate cases as well.

### References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 4.

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